

VeyaAI

*A Personal AI who understands your life,
so you don't have to figure it out alone.*

SUBMITTED FOR
Patchamomma 2026

CATEGORY
Personal Life Intelligence

PHASE
Phase 2 (Integration & Action)

SECTION 01: The Evolution

The Problem / Gap (Phase 2 Focus)

In Phase 1, we successfully proved that a Personal Digital Twin could observe a user's life and extract deep contextual meaning. However, we discovered a crucial gap: Insight without action creates anxiety.

Users do not want another dashboard telling them they are stressed. They want a system that negotiates their calendar, respects their limits, and actively reshuffles their life before burnout hits. Most tools stop at analytics; VeyaAI Phase 2 crosses the threshold into execution.

The 7-Dimensional Cognitive Decision Matrix

To act on behalf of a human, VeyaAI uses a sophisticated 7-D mathematical engine to weigh tradeoffs, prioritizing what matters most:

- 1. Chrono-Energy: Accounting for peak focus hours and transition lag after hard meetings.
- 2. Relational Capital: Dynamic 'Household Orbit' mapping, weighting the importance (1-100%) of family members to prevent relational deficit.
- 3. Cognitive Load: Distinguishing high-stakes preparation from routine execution.
- 4. Biological Reserves: Factor in sleep debt and physiological rhythms.
- 5. Physical Geography: Transit buffers and spatial displacement.
- 6. Psychological Resilience: Voice-intervention fallback triggers when high anxiety is detected.
- 7. Downstream Costs: Measuring the compounding penalty of delaying an event.

SECTION 02: Data & The Feedback Loop

Enriched Real-World Data Sources

Phase 2 deepens Veya's intelligence by connecting to robust, real-world data streams:

- Household Orbit Profile (New): Explicit 1-100% importance weighting for key people in the user's life, driving relational prioritization.
- Health Connect / Wearables: Integrating real-time sleep debt and physiological stress markers directly into the Cognitive Matrix.
- Live Google Calendar API: Moving beyond read-only to bi-directional sync (Take-Action-On-Approval).

The Maturity Engine (Feedback Loop)

VeyaAI gets smarter through telemetry. When Veya proposes an 'Action Card' (e.g., 'Compress meeting to 15 mins'), and the user applies it, the action executes via API. A micro-survey later asks if the intervention helped. This feedback loop is stored in the 'action_outcomes' database, filtering noise and allowing Veya to eventually auto-execute high-confidence patterns.

SECTION 03: Technology & Multi-Agent Architecture

6-Agent Orchestration Pipeline

VeyaAI's intelligence is distributed across highly specialized, deterministic agents routed by an Orchestrator:

- 1. OnboardingEngine: Conversational intercept to gather baseline anchors gently.
- 2. ContextAgent: Aggregates Live Calendar APIs with the local SQLite Household Orbit profiles.
- 3. TwinBuilder: Extracts and writes static facts to long-term vector memory.
- 4. PatternAgent: The math engine powering the 7-D Cognitive Matrix.
- 5. ActionAgent: The Negotiator. Computes realistic, 1-tap JSON Action Cards.
- 6. RagAgent / GeminiEngine: The voice. Uses TF-IDF cosine similarity to pull past relevant memories and synthesize an empathetic reply via Vertex AI.

Frontend Interventions

The Flutter-inspired web client brings the intelligence to life:

- Tomorrow Planner: A split view contrasting the user's rigid existing calendar against Veya's flexible suggestions--actionable blocks optimized for cognitive focus.
- Actionable Insights: Translating static stress scores into contextual nudges (e.g. 'Step onto the balcony') with a 1-tap 'Nudge Me' reminder integration.
- Voice Fallback: A real-time anxiety detection system that interrupts text chat to offer a live audio-call intervention.